Grade 10 – Term 1
Decision Trees with Staged Filtering: C9 Practice

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Evaluated criteria

C9 Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

1 Introduction

Use the supplied ordered criterion results to apply the filtering rules in order. At each stage, compare only the alternatives retained by the previous stage. Higher EV and Maximin values are preferred; smaller maximum regrets are preferred. Keep the supplied regret values fixed while filtering.

Represent the process as labelled filtering nodes showing the retained alternatives and final selected leaf. These nodes describe successive comparisons of alternatives. State probabilities belong to the payoff model used to calculate EV.



2 Decision Trees with Staged Filtering

Problems:

2.1. City logistics policy

2.2. Regional distributor expansion

2.3. Trade finance guarantees

2.4. Insurance portfolio mixes

2.1 City Logistics Policy

Problem Description

A city logistics office must choose one policy among $B_1, B_2, B_3, B_4, B_5, B_6$. Payoffs (thousand USD) and probabilities are:

State	Probability	B_1	B_2	B_3	B_4	B_5	B_6
T_1	0.50	95	90	110	85	100	92
T_2	0.30	70	75	60	80	65	78
T_3	0.20	30	40	10	55	35	20

Ordered criterion results (already computed):

$$\text{EV ranking: } B_4(77.50) > B_5(76.50) > B_2(75.50) > B_3(75.00) > B_1(74.50) > B_6(73.40).$$

$$\text{Maximin ranking: } B_4(55) > B_2(40) > B_5(35) > B_1(30) > B_6(20) > B_3(10).$$

$$\text{Minimax Regret ranking: } B_2(20) = B_5(20) > B_1(25) = B_4(25) > B_6(35) > B_3(45).$$

Filtering rule: (1) keep the best 4 EV options; (2) from those, keep the best 2 Maximin options; (3) from those, choose the smallest Minimax Regret.

Questions/Tasks

C9 Apply the filtering rules using the supplied results. Construct a labelled decision tree showing the retained alternatives at each stage, and state the final decision.

Solution

Step 1. Reconfirm ordered lists.

- EV top 4: $\{B_4, B_5, B_2, B_3\}$.
- On this subset, Maximin values are $B_4(55),\ B_2(40),\ B_5(35),\ B_3(10)$, so top 2 are $\{B_4, B_2\}$.
- On $\{B_4, B_2\}$, Max regrets are $B_4(25),\ B_2(20)$.

Step 2. Decision tree for staged filtering.

Decision tree (filtering nodes and leaves)
Root decision node D_0: All alternatives $\{B_1, B_2, B_3, B_4, B_5, B_6\}$
$\rightarrow$ Node D_1 (EV filter): keep $\{B_4, B_5, B_2, B_3\}$
$\rightarrow$ Node D_2 (Maximin filter): keep $\{B_4, B_2\}$
$\rightarrow$ Node D_3 (Minimax Regret filter): choose smallest max regret
Leaves: B_2 (20), B_4 (25)
Final leaf selected: B_2

Step 3. Final decision. The decision tree selects B_2.

2.2 Regional Distributor Expansion

Problem Description

A regional distributor must choose one expansion plan among $X_1, X_2, X_3, X_4, X_5, X_6$. Payoffs (thousand USD) and probabilities are:

State	Probability	X_1	X_2	X_3	X_4	X_5	X_6
U_1	0.30	130	120	140	110	125	118
U_2	0.25	90	95	80	100	88	92
U_3	0.25	60	70	55	85	65	75
U_4	0.20	20	40	10	60	45	35

Ordered criterion results (already computed):

$$\text{EV ranking: } X_4(91.25) > X_2(85.25) > X_5(84.75) > X_6(84.15) > X_1(80.50) > X_3(77.75).$$

$$\text{Maximin ranking: } X_4(60) > X_5(45) > X_2(40) > X_6(35) > X_1(20) > X_3(10).$$

$$\text{Minimax Regret ranking: } X_2(20) = X_5(20) > X_6(25) > X_4(30) > X_1(40) > X_3(50).$$

Filtering rule: (1) keep the best 3 EV options; (2) from those, keep the best 2 Minimax Regret options; (3) from those, choose the best Maximin option.

Questions/Tasks



C9 Apply the filtering rules using the supplied results. Construct a labelled decision tree showing the retained alternatives at each stage, and state the final decision.

Solution

Step 1. Reconfirm ordered lists and filtering sets.

- EV top 3: $\{X_4, X_2, X_5\}$.
- In this subset, Minimax Regret values are $X_2(20), X_5(20), X_4(30)$, so keep $\{X_2, X_5\}$.
- In $\{X_2, X_5\}$, Maximin values are $X_2(40), X_5(45)$.

Step 2. Decision tree for staged filtering.

Decision tree (filtering nodes and leaves)
Root decision node D_0: $\{X_1, X_2, X_3, X_4, X_5, X_6\}$
$\rightarrow$ Node D_1 (EV filter): keep $\{X_4, X_2, X_5\}$
$\rightarrow$ Node D_2 (Minimax Regret filter): keep $\{X_2, X_5\}$
$\rightarrow$ Node D_3 (Maximin choice)
Leaves: X_2 (min $= 40$), X_5 (min $= 45$)
Final leaf selected: X_5

Step 3. Final decision. The decision tree selects X_5.

2.3 Trade Finance Guarantees

Problem Description

A trade finance agency chooses one guarantee policy among $G_1, G_2, G_3, G_4, G_5, G_6$. States and probabilities are shown below. Payoffs are in thousand USD. Policies G_4 and G_5 balance upside and downside because they diversify by sector, while G_3 is highly cyclical.

State	Probability	G_1	G_2	G_3	G_4	G_5	G_6
M_1	0.45	100	94	114	90	104	96
M_2	0.35	72	78	64	84	70	80
M_3	0.20	32	44	12	56	38	26

Ordered criterion results (already computed):

$$\text{EV ranking: } G_4(83.90) > G_5(81.70) > G_2(81.20) > G_3(79.70) > G_1(78.10) > G_6(75.40).$$

$$\text{Maximin ranking: } G_4(56) > G_2(44) > G_5(38) > G_1(32) > G_6(26) > G_3(12).$$

$$\text{Minimax Regret ranking: } G_2(20) = G_5(20) > G_4(24) > G_1(24) > G_6(34) > G_3(44).$$

Filtering rule: (1) keep best 4 EV options; (2) from those, keep best 2 Maximin options; (3) from those, choose smallest Minimax Regret.

Questions/Tasks

C9 Apply the filtering rules using the supplied results. Construct a labelled decision tree showing the retained alternatives at each stage, and state the final decision.

Solution

Step 1. Reconfirm ordered lists.

- EV top 4: $\{G_4, G_5, G_2, G_3\}$.
- On this subset, Maximin values are $G_4(56), G_2(44), G_5(38), G_3(12)$, so keep $\{G_4, G_2\}$.
- On $\{G_4, G_2\}$, max regrets are $G_4(24), G_2(20)$.

Step 2. Decision tree for staged filtering.

Decision tree (filtering nodes and leaves)
Root decision node D_0: all alternatives $\{G_1, G_2, G_3, G_4, G_5, G_6\}$
$\rightarrow$ Node D_1 (EV filter): keep $\{G_4, G_5, G_2, G_3\}$
$\rightarrow$ Node D_2 (Maximin filter): keep $\{G_4, G_2\}$
$\rightarrow$ Node D_3 (Minimax Regret filter): choose smallest max regret
Leaves: G_2 (20), G_4 (24)
Final leaf selected: G_2

Step 3. Final decision. The decision tree selects G_2.



2.4 Insurance Portfolio Mixes

Problem Description

An insurance broker considers six portfolio mixes Y_1 to Y_6 under four climate-risk states with probabilities W_1:0.30, W_2:0.25, W_3:0.25, W_4:0.20. Payoffs are in thousand USD. Y_4 is conservative and protects downside in severe-weather years, while Y_3 has the highest upside but a weak stress outcome.

State	Probability	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6
W_1	0.30	136	126	146	116	130	121
W_2	0.25	94	98	84	103	90	95
W_3	0.25	66	74	58	88	69	77
W_4	0.20	28	46	14	64	48	38

Ordered criterion results (already computed):

$$\text{EV ranking: } Y_4(95.25) > Y_2(89.20) > Y_5(88.70) > Y_6(87.95) > Y_1(84.90) > Y_3(82.20).$$

$$\text{Maximin ranking: } Y_4(64) > Y_5(48) > Y_2(46) > Y_6(38) > Y_1(28) > Y_3(14).$$

$$\text{Minimax Regret ranking: } Y_2(20) = Y_5(20) > Y_6(26) > Y_4(30) > Y_1(40) > Y_3(50).$$

Filtering rule: (1) keep the best 3 EV options; (2) from those, keep the best 2 Minimax Regret options; (3) from those, choose the best Maximin option.

Questions/Tasks



C9 Apply the filtering rules using the supplied results. Construct a labelled decision tree showing the retained alternatives at each stage, and state the final decision.

Solution

Step 1. Reconfirm ordered lists and filtering sets.

- EV top 3: $\{Y_4, Y_2, Y_5\}$.
- In this subset, Minimax Regret values are $Y_2(20), Y_5(20), Y_4(30)$, so keep $\{Y_2, Y_5\}$.
- In $\{Y_2, Y_5\}$, Maximin values are $Y_2(46), Y_5(48)$.

Step 2. Decision tree for staged filtering.

Decision tree (filtering nodes and leaves)
Root decision node D_0: $\{Y_1, Y_2, Y_3, Y_4, Y_5, Y_6\}$
$\rightarrow$ Node D_1 (EV filter): keep $\{Y_4, Y_2, Y_5\}$
$\rightarrow$ Node D_2 (Minimax Regret filter): keep $\{Y_2, Y_5\}$
$\rightarrow$ Node D_3 (Maximin choice)
Leaves: Y_2 (min $= 46$), Y_5 (min $= 48$)
Final leaf selected: Y_5

Step 3. Final decision. The decision tree selects Y_5.

